

6-7b

1. If $G(x)$ is an antiderivative for $f(x)$ and $G(2) = -7$, then $G(4) =$

(A) $f'(4)$

(B) $-7 + f'(4)$

(C) $\int_2^4 f(t) dt$

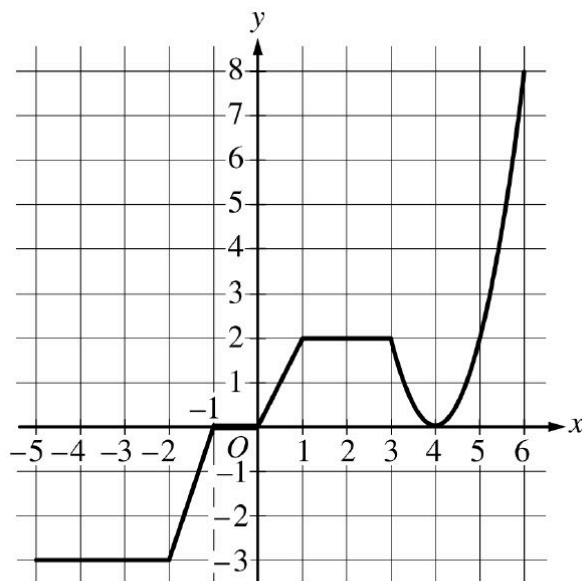
(D) $\int_2^4 (-7 + f(t)) dt$

(E) $-7 + \int_2^4 f(t) dt$



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2.

Graph of g

The graph of the continuous function g , the derivative of the function f , is shown above. The function g is piecewise linear for $-5 \leq x < 3$, and $g(x) = 2(x - 4)^2$ for $3 \leq x \leq 6$.

(a) If $f(1) = 3$, what is the value of $f(-5)$?

(b) Evaluate $\int_1^6 g(x) dx$.

(c) For $-5 < x < 6$, on what open intervals, if any, is the graph of f both increasing and concave up? Give a reason for your answer.

(d) Find the x -coordinate of each point of inflection of the graph of f . Give a reason for your answer.

Part A

1 point is earned for integral

1 point is earned for answer

$$\begin{aligned}
 f(-5) &= f(1) + \int_1^{-5} g(x) dx = f(1) - \int_{-5}^1 g(x) dx \\
 &= 3 - \left(-9 - \frac{3}{2} + 1\right) = 3 - \left(-\frac{19}{2}\right) = \frac{25}{2}
 \end{aligned}$$



0	1	2
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The student response earns all of the following points:

1 point is earned for integral

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1 point is earned for answer

$$\begin{aligned} f(-5) &= f(1) + \int_1^{-5} g(x)dx = f(1) - \int_{-5}^1 g(x)dx \\ &= 3 - \left(-9 - \frac{3}{2} + 1\right) = 3 - \left(-\frac{19}{2}\right) = \frac{25}{2} \end{aligned}$$

Part B

1 point is earned for split at $x = 3$

1 point is earned for antiderivative $2(x - 4)^2$

1 point is earned for answer

$$\begin{aligned} \int_1^6 g(x)dx &= \int_1^3 g(x)dx + \int_3^6 g(x)dx \\ &= \int_1^3 2dx + \int_3^6 2(x - 4)^2dx \\ &= 4 + \left[\frac{2}{3}(x - 4)^3\right]_{x=3}^{x=6} = 4 + \frac{16}{3} - \left(-\frac{2}{3}\right) = 10 \end{aligned}$$



0	1	2	3
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1 point is earned for split at $x = 3$

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Part C

1 point is earned for intervals

1 point is earned for reason

The graph of f is increasing and concave up on $0 < x < 1$ and $4 < x < 6$ because $f'(x) = g(x) > 0$ and $f'(x) = g(x)$ is increasing on those intervals.



0	1	2
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The student response earns all of the following points:

1 point is earned for intervals

1 point is earned for reason

The graph of f is increasing and concave up on $0 < x < 1$ and $4 < x < 6$ because $f'(x) = g(x) > 0$ and $f'(x) = g(x)$ is increasing on those intervals.

Part D

1 point is earned for answer

1 point is earned for reason

The graph of f has a point of inflection at $x = 4$ because $f'(x) = g(x)$ changes from decreasing to increasing at $x = 4$.



0	1	2
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The student response earns all of the following points:

1 point is earned for answer

1 point is earned for reason

The graph of f has a point of inflection at $x = 4$ because $f'(x) = g(x)$ changes from decreasing to increasing at $x = 4$.

6-7b

3. A cubic polynomial function f is defined by

$$f(x) = 4x^3 + ax^2 + bx + k$$

where a , b , and k are constants. The function f has a local minimum at $x = -1$, and the graph f has a point of inflection at $x = -2$.

(a) Find the values of a and b .

(b) If $\int_0^1 f(x) dx = 32$, what is the value of k ?

Part A

1 point is earned for: $f'(x)$

1 point is earned for: $f''(x)$

1 point is earned for: $f'(-1) = 0$

1 point is earned for: $f''(-2) = 0$

1 point is earned for: a , b

$$f'(x) = 12x^2 + 2ax + b$$

$$f''(x) = 24x + 2a$$

$$f'(-1) = 12 - 2a + b = 0$$

$$f''(-2) = -48 + 2a = 0$$

$$a = 24$$

$$b = -12 + 2a = 36$$



0	1	2	3	4	5
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The student response earns all of the following points:

1 point is earned for: $f'(x)$

1 point is earned for: $f''(x)$

1 point is earned for: $f'(-1) = 0$

1 point is earned for: $f''(-2) = 0$

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1 point is earned for: a, b

$$f'(x) = 12x^2 + 2ax + b$$

$$f''(x) = 24x + 2a$$

$$f'(-1) = 12 - 2a + b = 0$$

$$f''(-2) = -48 + 2a = 0$$

$$a = 24$$

$$b = -12 + 2a = 36$$

Part B

2 points are earned for antidifferentiation <-1> each error

1 point is earned for the expression in k

1 point is earned for k

$$\int_0^1 (4x^3 + 24x^2 + 36x + k) dx$$

$$= x^4 + 8x^3 + 18x^2 + kx \Big|_{x=0}^{x=1} = 27 + k$$

$$27 + k = 32$$

$$k = 5$$



0	1	2	3	4
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The student response earns all of the following points:

2 points are earned for antidifferentiation <-1> each error

1 point is earned for the expression in k

1 point is earned for k

6-7b

$$\begin{aligned}\int_0^1 (4x^3 + 24x^2 + 36x + k) dx \\&= x^4 + 8x^3 + 18x^2 + kx \Big|_{x=0}^{x=1} = 27 + k \\27 + k &= 32 \\k &= 5\end{aligned}$$

6-7b

4. A cubic polynomial function f is defined by

$$f(x) = 4x^3 + ax^2 + bx + k$$

where a , b , and k are constants. The function f has a local minimum at $x = -1$, and the graph f has a point of inflection at $x = -2$.

If $\int_0^1 f(x)dx = 32$, what is the value of k ?

Part B

2 points are earned for antidifferentiation < -1 > each error

1 point is earned for the expression in k

1 point is earned for k

$$\begin{aligned} \int_0^1 (4x^3 + 24x^2 + 36x + k)dx \\ = x^4 + 8x^3 + 18x^2 + kx \Big|_{x=0}^{x=1} = 27 + k \\ 27 + k = 32 \\ k = 5 \end{aligned}$$



0	1	2	3	4
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The student response earns all of the following points:

2 points are earned for antidifferentiation < -1 > each error

1 point is earned for the expression in k

1 point is earned for k

$$\begin{aligned} \int_0^1 (4x^3 + 24x^2 + 36x + k)dx \\ = x^4 + 8x^3 + 18x^2 + kx \Big|_{x=0}^{x=1} = 27 + k \\ 27 + k = 32 \\ k = 5 \end{aligned}$$

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5.

x	-1.5	-1.0	-0.5	0	0.5	1.0	1.5
$f(x)$	-1	-4	-6	-7	-6	-4	-1
$f'(x)$	-7	-5	-3	0	3	5	7

Let f be a function that is differentiable for all real numbers. The table above gives the values of f and its derivative f' for selected points x in the closed interval $-1.5 \leq x \leq 1.5$. The second derivative of f has the property that $f''(x) > 0$ for $-1.5 \leq x \leq 1.5$.

Evaluate $\int_0^{1.5} (3f'(x) + 4) dx$. Show the work that leads to your answer.

Part A

1 point is earned for the antiderivative

1 point is earned for the answer

$$\begin{aligned} \int_0^{1.5} (3f'(x) + 4) dx &= 3 \int_0^{1.5} 4 dx \\ &= 3f(x) + 4x \Big|_0^{1.5} = 3(-1 - (-7)) + 4(1.5) = 24 \end{aligned}$$



0	1	2
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The student response earns all of the following points:

1 point is earned for the antiderivative

1 point is earned for the answer

$$\begin{aligned} \int_0^{1.5} (3f'(x) + 4) dx &= 3 \int_0^{1.5} 4 dx \\ &= 3f(x) + 4x \Big|_0^{1.5} = 3(-1 - (-7)) + 4(1.5) = 24 \end{aligned}$$

6-7b

6. The function g is defined for $x > 0$ with $g(1)=2$, $g'(x)=\sin(x+1/x)$, and $g''(x)=(1-1/x^2)\cos(x+1/x)$.



Write an equation for the line tangent to the graph of g at $x=0.3$.

Part C

1 point is earned for $g'(0.3)$

1 point is earned for the integral expression

1 point is earned for $g(0.3)$

1 point is earned for the equation

$$g'(0.3) = -0.472161$$

$$g(0.3) = 2 + \int_1^{0.3} g'(x) dx = 1.546007$$

An equation for the line tangent to the graph of g is $y = 1.546 - 0.472(x - 0.3)$.



0	1	2	3	4
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$$g'(0.3) = -0.472161$$

$$g(0.3) = 2 + \int_1^{0.3} g'(x) dx = 1.546007$$

An equation for the line tangent to the graph of g is

$$y = 1.546 - 0.472(x - 0.3).$$

6-7b

7. Consider a differentiable function f having domain all positive real numbers, and for which it is known that $f'(x) = (4 - x)x^{-3}$ for $x > 0$.
- (a) Find the x -coordinate of the critical point of f . Determine whether the point is a relative maximum, a relative minimum, or neither for the function f . Justify your answer.
- (b) Find all intervals on which the graph of f is concave down. Justify your answer.
- (c) Given that $f(1) = 2$, determine the function f .

Part A

1 point is earned for $x = 4$

1 point is earned for relative maximum

1 point is earned for the justification

$$f'(x) = 0 \text{ at } x = 4$$

$$f'(x) > 0 \text{ for } 0 < x < 4$$

$$f'(x) < 0 \text{ for } x > 4$$

Therefore f has a relative maximum at $x = 4$.



0	1	2	3
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The student response earns all of the following points:

1 point is earned for $x = 4$

1 point is earned for relative maximum

1 point is earned for the justification

$$f'(x) = 0 \text{ at } x = 4$$

$$f'(x) > 0 \text{ for } 0 < x < 4$$

$$f'(x) < 0 \text{ for } x > 4$$

Therefore f has a relative maximum at $x = 4$.

Part B

2 points are earned for $f''(x)$

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1 point is earned for answer with justification

$$\begin{aligned}
 f''(x) &= -x^{-3} + (4 - x)(-3x^{-4}) \\
 &= -x^{-3} - 12x^{-4} + 3x^{-3} \\
 &= 2x^{-4}(x - 6) \\
 &= \frac{2(x-6)}{x^4}
 \end{aligned}$$

$$f''(x) < 0 \text{ for } 0 < x < 6$$

The graph of f is concave down on the interval $0 < x < 6$.



0	1	2	3
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The student response earns all of the following points:

2 points are earned for $f''(x)$

1 point is earned for answer with justification

$$\begin{aligned}
 f''(x) &= -x^{-3} + (4 - x)(-3x^{-4}) \\
 &= -x^{-3} - 12x^{-4} + 3x^{-3} \\
 &= 2x^{-4}(x - 6) \\
 &= \frac{2(x-6)}{x^4}
 \end{aligned}$$

$$f''(x) < 0 \text{ for } 0 < x < 6$$

The graph of f is concave down on the interval $0 < x < 6$.

Part C

1 point is earned for integral

1 point is earned for antiderivative

1 point is earned for the answer

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$$\begin{aligned}f(x) &= 2 + \int_1^x (4t^{-3} - t^{-2}) dt \\&= 2 + \left[-2t^{-2} + t^{-1}\right]_{t=1}^{t=x} \\&= 3 - 2x^{-2} + x^{-1}\end{aligned}$$



0	1	2	3
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The student response earns all of the following points:

1 point is earned for integral

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1 point is earned for the answer

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6-7b

8. Consider a differentiable function f having domain all positive real numbers, and for which it is known that $f'(x) = (4-x)x^{-3}$ for $x > 0$.

Given that $f(1) = 2$, determine the function f

Part C

1 point is earned for integral

1 point is earned for antiderivative

1 point is earned for answer

$$f(x) = 2 + \int_1^x (4t^{-3} - t^{-2}) dt$$

$$= 2 + [-2t^{-2} + t^{-1}]_{t=1}^{t=x}$$

$$= 3 - 2x^{-2} + x^{-1}$$



0	1	2	3
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The student response earns all of the following points:

1 point is earned for integral

1 point is earned for antiderivative

1 point is earned for answer

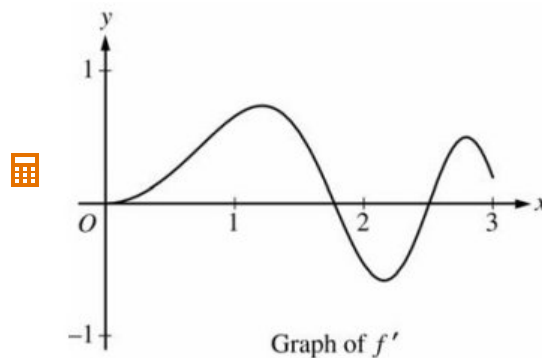
$$f(x) = 2 + \int_1^x (4t^{-3} - t^{-2}) dt$$

$$= 2 + [-2t^{-2} + t^{-1}]_{t=1}^{t=x}$$

$$= 3 - 2x^{-2} + x^{-1}$$

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9.



Let f be the function defined for $x \geq 0$ with $f(0) = 5$ and f' , the first derivative of f , given by $f'(x) = e^{(-x/4)} \sin(x^2)$. The graph of $y = f'(x)$ is shown above.

(a) Use the graph of f' to determine whether the graph of f is concave up, concave down, or neither on the interval $1.7 < x < 1.9$. Explain your reasoning.

(b) On the interval $0 \leq x \leq 3$, find the value of x at which f has an absolute maximum. Justify your answer.

(c) Write an equation for the line tangent to the graph of f at $x = 2$.

Part A

1 point is earned for the answer

1 point is earned for the reason

On the interval $1.7 < x < 1.9$, f' is decreasing and thus f is concave down on this interval.



0	1	2
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The student response earns all of the following points:

1 point is earned for the answer

1 point is earned for the reason

On the interval $1.7 < x < 1.9$, f' is decreasing and thus f is concave down on this interval.

Part B

1 point is earned if identified $\sqrt{\pi}$ and 3 as candidates **or** indicates that the graph of f increases, decreases, then increases

1 point is earned if justified $f(\sqrt{\pi}) > f(3)$

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1 point is earned for answer

$$f'(x) = 0 \text{ when } x = 0, \sqrt{\pi}, \sqrt{2\pi}, \sqrt{3\pi}, \dots$$

On $[0, 3]$, f' changes from positive to negative only at $\sqrt{\pi}$. The absolute maximum must occur at $x = \sqrt{\pi}$ or at an endpoint.

$$f(0) = 5$$

$$f(\sqrt{\pi}) = f(0) + \int_0^{\sqrt{\pi}} f'(x) dx = 5.67911$$

$$f(3) = f(0) + \int_0^3 f'(x) dx = 5.57893$$

This shows that f has an absolute maximum at $x = \sqrt{\pi}$.



0	1	2	3
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The student response earns all of the following points:

1 point is earned if identified $\sqrt{\pi}$ and 3 as candidates **or** indicates that the graph of f increases, decreases, then increases

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$$f(0) = 5$$

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$$f(3) = f(0) + \int_0^3 f'(x) dx = 5.57893$$

This shows that f has an absolute maximum at $x = \sqrt{\pi}$.

Part C

1 point is earned for $f(2)$ expression with integral

1 point is earned for $f(2)$ expression including $f(0)$ term

1 point is earned for $f'(2)$

1 point is earned for equation

$$f(2) = f(0) + \int_0^2 f'(x) dx = 5.62342$$

$$f'(2) = e^{-0.5} \sin(4) = -0.45902$$

$$y - 5.623 = (-0.459)(x - 2)$$



0	1	2	3	4
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1 point is earned for $f(2)$ expression including $f(0)$ term

1 point is earned for $f'(2)$

1 point is earned for equation

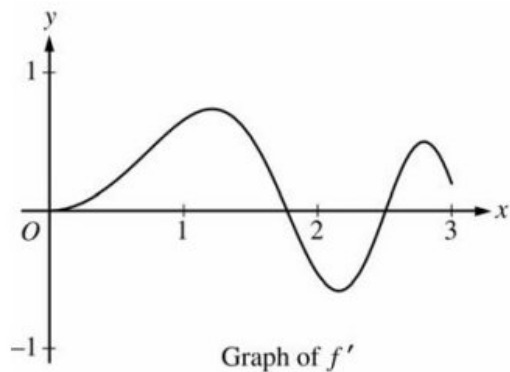
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$$f'(2) = e^{-0.5} \sin(4) = -0.45902$$

$$y - 5.623 = (-0.459)(x - 2)$$

6-7b

6-7b



Let f be the function defined for $x \geq 0$ with $f(0)=5$ and f' , the first derivative of f , given by $f'(x)=e^{(-x/4)}\sin(x^2)$. The graph of $y=f'(x)$ is shown above.

10.  On the interval $0 \leq x \leq 3$, find the value of x at which f has an absolute maximum. Justify your answer.

FRQ Score

1 point is earned if identified $\sqrt{\pi}$ and 3 as candidates **or** indicates that the graph of f increases, decreases, then increases

1 point is earned if justified $f(\sqrt{\pi}) > f(3)$

1 point is earned for answer

$$f'(x)=0 \text{ when } x=0, \sqrt{\pi}, \sqrt{2\pi}, \sqrt{3\pi}, \dots$$

On $[0, 3]$ f' changes from positive to negative only at $\sqrt{\pi}$. The absolute maximum must occur at $x = \sqrt{\pi}$ or at an endpoint.

$$f(0)=5$$

$$f(\sqrt{\pi}) = f(0) + \int_0^{\sqrt{\pi}} f'(x) dx = 5.67911$$

$$f(3) = f(0) + \int_0^3 f'(x) dx = 5.57893$$

This shows that f has an absolute maximum at $x = \sqrt{\pi}$

6-7b



0	1	2	3
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The student response earns all of the following points:

1 point is earned if identified $\sqrt{\pi}$ and 3 as candidates **or** indicates that the graph of f increases, decreases, then increases

1 point is earned if justified $f(\sqrt{\pi}) > f(3)$

1 point is earned for answer

$f'(x)=0$ when $x=0$,

On $[0, 3]$ f' changes from positive to negative only at $\sqrt{\pi}$. The absolute maximum must occur at $x = \sqrt{\pi}$ or at an endpoint.

$f(0)=5$

$$f(\sqrt{\pi}) = f(0) + \int_0^{\sqrt{\pi}} f'(x)dx = 5.67911$$

$$f(3) = f(0) + \int_0^3 f'(x)dx = 5.57893$$

This shows that f has an absolute maximum at $x = \sqrt{\pi}$

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11.  Write an equation for the line tangent to the graph of f at $x=2$.

Part C

1 point is earned for $f(2)$ expression with integral

1 point is earned for $f(2)$ expression including $f(0)$ term

1 point is earned for $f'(2)$

1 point is earned for equation

$$f(2)=f(0)+\int_0^2 f'(x)dx = 5.62342$$

$$f'(2)=e-0.5\sin(4) = -0.45902$$

$$y-5.623 = (-0.459)(x-2)$$



0	1	2	3	4
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The student response earns all of the following points:

1 point is earned for $f(2)$ expression with integral

1 point is earned for $f(2)$ expression including $f(0)$ term

1 point is earned for $f'(2)$

1 point is earned for equation

$$f(2)=f(0)+\int_0^2 f'(x)dx = 5.62342$$

$$f'(2)=e-0.5\sin(4) = -0.45902$$

$$y-5.623 = (-0.459)(x-2)$$

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Let R be the region enclosed by the graphs of $y=e^x$, $y=(x-1)^2$, and the line $x=1$.

12. Find the area of R .

Part A

1 point is earned for definite integral

1 point is **deducted** for incorrect limits

1 point is **deducted** for reversal

1 point is earned for anti derivative

1 point is earned for evaluation

$$\begin{aligned}
 A &= \int_0^1 e^x - (x-1)^2 dx \\
 &= \int_0^1 e^x - x^2 + 2x - 1 dx \\
 &= e^x \Big|_0^1 - \frac{1}{3}(x-1)^3 \Big|_0^1 \\
 &= (e-1) - \frac{1}{3} = e - \frac{4}{3}
 \end{aligned}$$



0	1	2	3
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The student response earns three of the following points:

1 point is earned for definite integral

1 point is **deducted** for incorrect limits

1 point is **deducted** for reversal

1 point is earned for anti derivative

1 point is earned for evaluation

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$$\begin{aligned} A &= \int_0^1 e^x - (x-1)^2 dx \\ &= \int_0^1 e^x - x^2 + 2x - 1 dx \\ &= e^x \Big|_0^1 - \frac{1}{3}(x-1)^3 \Big|_0^1 \\ &= (e-1) - \frac{1}{3} = e - \frac{4}{3} \end{aligned}$$

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13. Find the volume of the solid generated when R is revolved about the x -axis.

Part B

2 points are earned for definite integral

2 points are **deducted** for faulty f , g , or u

1 point is earned for correct antiderivative

1 point is earned for evaluation of anti derivative

$$\begin{aligned}
 V &= \pi \int_0^1 e^{2x} - (x-1)^4 dx \\
 &= \pi \left[\frac{e^{2x}}{2} \right]_0^1 - \pi \left[\frac{1}{5}(x-1)^5 \right] \Big|_0^1 \\
 &= \pi \left[\left(\frac{e^2}{2} - \frac{1}{2} \right) - \frac{1}{5} \right] = \pi \left(\frac{e^2}{2} - \frac{7}{10} \right)
 \end{aligned}$$

or

$$\begin{aligned}
 V &= 2\pi \int_0^1 y[1 - (1 - \sqrt{y})]dy + 2\pi \int_1^e y(1 - \ln y)dy \\
 &= 2\pi \cdot \frac{2}{5}y^{5/2} \Big|_0^1 + 2\pi \left[\frac{1}{2}y^2 - \left(\frac{1}{2}y^2 \ln y - \frac{1}{4}y^2 \right) \right] \Big|_1^e \\
 &= \frac{4}{5}\pi + 2\pi \left[\frac{1}{4}e^2 - \frac{3}{4} \right] = \pi \left(\frac{e^2}{2} - \frac{7}{10} \right)
 \end{aligned}$$



0	1	2	3
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The student response earns three of the following points:

2 points are earned for definite integral

2 points are **deducted** for faulty f , g , or u

1 point is earned for correct antiderivative

1 point is earned for evaluation of anti derivative

6-7b

$$\begin{aligned} V &= \pi \int_0^1 e^{2x} - (x-1)^4 dx \\ &= \pi \left[\frac{e^{2x}}{2} \right]_0^1 - \pi \left[\frac{1}{5}(x-1)^5 \right] \bigg|_0^1 \\ &= \pi \left[\left(\frac{e^2}{2} - \frac{1}{2} \right) - \frac{1}{5} \right] = \pi \left(\frac{e^2}{2} - \frac{7}{10} \right) \end{aligned}$$

or

$$\begin{aligned} V &= 2\pi \int_0^1 y[1 - (1 - \sqrt{y})] dy + 2\pi \int_1^e y(1 - \ln y) dy \\ &= 2\pi \cdot \frac{2}{5} y^{5/2} \bigg|_0^1 + 2\pi \left[\frac{1}{2} y^2 - \left(\frac{1}{2} y^2 \ln y - \frac{1}{4} y^2 \right) \right] \bigg|_1^e \\ &= \frac{4}{5} \pi + 2\pi \left[\frac{1}{4} e^2 - \frac{3}{4} \right] = \pi \left(\frac{e^2}{2} - \frac{7}{10} \right) \end{aligned}$$

6-7b

14. Let R be the region in the first quadrant under the graph of $y = \frac{x}{x^2+2}$ for $0 \leq x \leq \sqrt{6}$.
If the line $x = k$ divides R into two regions of equal area, what is the value of k ?

Part B

1 point is earned for a correct equation involving integral(s)

1 point is earned for antiderivative

1 point is earned for evaluation of integral(s)

1 point is earned for finding the value of k

$$\begin{aligned}\frac{1}{2} \ln &= \int_0^k \frac{x}{x^2+2} dx \\ &= \frac{1}{2} \ln(x^2+2) \Big|_0^k \\ &= \frac{1}{2} \ln(k^2+2) - \frac{1}{2} \ln 2 \\ \therefore \frac{1}{2} \ln(k^2+2) &= \frac{1}{2} \ln 2 + \frac{1}{2} \ln 2\end{aligned}$$

$$\text{or } \ln(k^2+2) = \ln 4$$

$$\therefore k^2+2 = 4 \text{ and } k = \sqrt{2}$$



0	1	2	3	4
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The student response earns four of the following points:

1 point is earned for a correct equation involving integral(s)

1 point is earned for antiderivative

1 point is earned for evaluation of integral(s)

1 point is earned for finding the value of k

6-7b

$$\begin{aligned}\frac{1}{2} \ln &= \int_0^k \frac{x}{x^2 + 2} dx \\&= \frac{1}{2} \ln(x^2 + 2) \Big|_0^k \\&= \frac{1}{2} \ln(k^2 + 2) - \frac{1}{2} \ln 2 \\ \therefore \frac{1}{2} \ln(k^2 + 2) &= \frac{1}{2} \ln 2 + \frac{1}{2} \ln 2\end{aligned}$$

$$\text{or } \ln(k^2 + 2) = \ln 4$$

$$\therefore k^2 + 2 = 4 \text{ and } k = \sqrt{2}$$

6-7b

15. Let $F(x) = \int_0^x \sin(t^2) dt$ for $0 \leq x \leq 3$.

If the average rate of change of F on the closed interval $[1,3]$ is k , find $\int_1^3 \sin(t^2) dt$ in terms of k .

Part C

1 point is earned for correctly solving $k = \frac{F(3)-F(1)}{2}$

$$k = \frac{F(3)-F(1)}{2} = \frac{\int_1^3 \sin(t^2) dt}{2}$$

1 point is earned for the correct answer

$$\int_1^3 \sin(t^2) dt = 2k$$



0	1	2
---	---	---

The student response earns two of the following points:

1 point is earned for correctly solving $k = \frac{F(3)-F(1)}{2}$

$$k = \frac{F(3)-F(1)}{2} = \frac{\int_1^3 \sin(t^2) dt}{2}$$

1 point is earned for the correct answer

$$\int_1^3 \sin(t^2) dt = 2k$$

16.  If the function f is defined by $f(x) = \sqrt{x^3 + 2}$ and g is an antiderivative of f such that $g(3) = 5$, then $g(1) =$

(A) -3.268

(B) -1.585

(C) 1.732

(D) 6.585

(E) 11.585



6-7b

17.  Let g be the function given by $g(x) = \int_1^x 100(t^2 - 3t + 2)e^{-t^2} dt$. Which of the following statements about g must be true?

- I. g is increasing on $(1, 2)$.
- II. g is increasing on $(2, 3)$.
- III. $g(3) > 0$

(A) I only

(B) II only



(C) III only

(D) II and III only

(E) I, II, and III

18. For $t \geq 0$, the position of a particle moving along the x -axis is given by $x(t) = \sin t - \cos t$. What is the acceleration of the particle at the point where the velocity is first equal to 0?

(A) $-\sqrt{2}$



(B) -1

(C) 0

(D) 1

(E) $\sqrt{2}$

6-7b

19.

t (minutes)	0	4	9	15	20
$W(t)$ (degrees Fahrenheit)	55.0	57.1	61.8	67.9	71.0

The temperature of water in a tub at time t is modeled by a strictly increasing, twice-differentiable function W , where $W(t)$ is measured in degrees Fahrenheit and t is measured in minutes. At time $t=0$, the temperature of the water is 55°F . The water is heated for 30 minutes, beginning at time $t=0$. Values of $W(t)$ at selected times t for the first 20 minutes are given in the table above.

Use the data in the table to evaluate $\int_0^{20} W'(t)dt$. Using correct units, interpret the meaning of $\int_0^{20} W'(t)dt$ in the context of this problem.

Part B

1 point is earned for the value

1 point is earned for the interpretation with units

$$\int_0^{20} W'(t)dt = W(20) - W(0) = 71.0 - 55.0 = 16$$

The water has warmed by 16 oF over the interval from $t=0$ to $t=20$ minutes.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the value

1 point is earned for the interpretation with units

$$\int_0^{20} W'(t)dt = W(20) - W(0) = 71.0 - 55.0 = 16$$

The water has warmed by 16 0F over the interval from $t=0$ to $t=20$ minutes.

6-7b

20. At a certain height, a tree trunk has a circular cross section. The radius $R(t)$ of that cross section grows at a rate modeled by the function

$$\frac{dR}{dt} = \frac{1}{16}(3 + \sin(t^2)) \text{ centimeters per year for}$$

$0 \leq t \leq 3$, where time t is measured in years. At time $t = 0$, the radius is 6 centimeters. The area of the cross section at time t is denoted by $A(t)$.

 Evaluate $\int_0^3 A'(t) dt$. Using appropriate units, interpret the meaning of that integral in terms of cross-sectional area.

Part C

1 point is earned for using Fundamental Theorem of Calculus

1 point is earned for value of $\int_0^3 A'(t) dt$.

1 point is earned for meaning of $\int_0^3 A'(t) dt$.

$$\int_0^3 A'(t) dt = A(3) - A(0) = 24.200 \text{ or } 24.201$$

From time $t=0$ to $t=3$ years, the cross-sectional area grows by 24.201 square centimeters.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for using Fundamental Theorem of Calculus

6-7b

1 point is earned for value of $\int_0^3 A'(t) \, dt$.

1 point is earned for meaning of $\int_0^3 A'(t) \, dt$.

$$\int_0^3 A'(t) \, dt = A(3) - A(0) = 24.200 \text{ or } 24.201$$

From time $t=0$ to $t=3$ years, the cross-sectional area grows by 24.201 square centimeters.

6-7b

21. The rate at which people enter an auditorium for a rock concert is modeled by the function R given by $R(t) = 1380t^2 - 675t^3$ for $0 \leq t \leq 2$ hours; $R(t)$ is measured in people per hour. No one is in the auditorium at time $t = 0$, when the doors open. The doors close and the concert begins at time $t = 2$.



The total wait time for all the people in the auditorium is found by adding the time each person waits, starting at the time the person enters the auditorium and ending when the concert begins. The function w models the total wait time for all the people who enter the auditorium before time t . The derivative of w is given by $w'(t) = (2-t)R(t)$. Find $w(2) - w(1)$, the total wait time for those who enter the auditorium after time $t = 1$.

Part C

1 point is earned for integral

1 point is earned for answer

$$w(2) - w(1) = \int_1^2 w'(t) dt = \int_1^2 (2-t)R(t) d(t) = 387.5$$

The total wait time for those who enter the auditorium after time $t = 1$ is 387.5 hours.



0	1	2
---	---	---

The student response earns two of the following points:

1 point is earned for integral

1 point is earned for answer

$$w(2) - w(1) = \int_1^2 w'(t) dt = \int_1^2 (2-t)R(t) d(t) = 387.5$$

The total wait time for those who enter the auditorium after time $t = 1$ is 387.5 hours.

6-7b

22.

Distance x (cm)	0	1	5	6	8
Temperature $T(x)$ ($^{\circ}\text{C}$)	100	93	70	62	55

A metal wire of length 8 centimeters (cm) is heated at one end. The table above gives selected values of the temperature $T(x)$, in degrees Celsius ($^{\circ}\text{C}$), of the wire x cm from the heated end. The function T is decreasing and twice differentiable.

Find $\int_0^8 T'(x)dx$, and indicate units of measure. Explain the meaning of $\int_0^8 T'(x)dx$ in terms of the temperature of the wire.

Part C

1 point is earned for the value

1 point is earned for the meaning

$$\int_0^8 T'(x)dx = T(8) - T(0) = 55 - 100 = -45^{\circ}\text{C}$$

The temperature drops 45°C from the heated end of the wire of the other end of the wire.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the value

1 point is earned for the meaning

$$\int_0^8 T'(x)dx = T(8) - T(0) = 55 - 100 = -45^{\circ}\text{C}$$

The temperature drops 45°C from the heated end of the wire of the other end of the wire.

6-7b

23.

t (minutes)	0	4	9	15	20
$W(t)$ (degrees Fahrenheit)	55.0	57.1	61.8	67.9	71.0

The temperature of water in a tub at time t is modeled by a strictly increasing, twice-differentiable function W , where $W(t)$ is measured in degrees Fahrenheit and t is measured in minutes. At time $t = 0$, the temperature of the water is 55°F . The water is heated for 30 minutes, beginning at time $t = 0$. Values of $W(t)$ at selected times t for the first 20 minutes are given in the table above.

 Use the data in the table to evaluate $\int_0^{20} W'(t) dt$. Using correct units, interpret the meaning of $\int_0^{20} W'(t) dt$ in the context of this problem.

Part B

1 point is earned for the value

$$\int_0^{20} w'(t) dt = w(20) - w(0) = 71.0 - 55.0 = 16$$

1 point is earned for the interpretation with units

The water has warmed by 16°F over the interval from $t=0$ to $t=20$ minutes.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the value

$$\int_0^{20} w'(t) dt = w(20) - w(0) = 71.0 - 55.0 = 16$$

1 point is earned for the interpretation with units

The water has warmed by 16°F over the interval from $t=0$ to $t=20$ minutes.

6-7b

24. Let R be the region in the first quadrant under the graph of $y = \frac{1}{\sqrt{x}}$ for $4 \leq x \leq 9$.

If the line $x=k$ divides the region R into two regions of equal area, what is the value of k ?

Part B

1 point is earned for the correct demonstration of variable limit using k as upper or lower limit with 4 or 9 (or imported limit)

$$\int_4^k \frac{dx}{\sqrt{x}} \text{ or } \int_k^9 \frac{dx}{\sqrt{x}}$$

1 point is earned for the correct equation involving the two halves of R that when solved will give answer with demonstrated algorithm that knows when to stop

1 point is earned for the correct answer

0/1 if answer from equation not involving relevant areas

1/3 bald answer

$$\int_4^k \frac{dx}{\sqrt{x}} = 1$$

$$2\sqrt{x} \Big|_4^k = 1$$

$$2\sqrt{k} - 2\sqrt{4} = 1$$

$$k = \frac{25}{4}$$

$$\left(\text{or } \int_k^9 \frac{dx}{\sqrt{x}} = 1 \text{ or } \int_4^k \frac{dx}{\sqrt{x}} = \int_k^9 \frac{dx}{\sqrt{x}} \right)$$



0	1	2	3
---	---	---	---

The student response earns three of the following points:

1 point is earned for the correct demonstration of variable limit using k as upper or lower limit with 4 or 9 (or imported limit)

$$\int_4^k \frac{dx}{\sqrt{x}} \text{ or } \int_k^9 \frac{dx}{\sqrt{x}}$$

1 point is earned for the correct equation involving the two halves of R that when solved will give answer with demonstrated algorithm that knows when to stop

1 point is earned for the correct answer

6-7b

0/1 if answer from equation not involving relevant areas

1/3 bald answer

$$\int_4^k \frac{dx}{\sqrt{x}} = 1$$

$$2\sqrt{x}]_4^k = 1$$

$$2\sqrt{k} - 2\sqrt{4} = 1$$

$$k = \frac{25}{4}$$

$$\left(\text{or } \int_k^9 \frac{dx}{\sqrt{x}} = 1 \text{ or } \int_4^k \frac{dx}{\sqrt{x}} = \int_k^9 \frac{dx}{\sqrt{x}} \right)$$

6-7b

25.

t (minutes)	0	2	5	9	10
$H(t)$ (degrees Celsius)	66	60	52	44	43



As a pot of tea cools, the temperature of the tea is modeled by a differentiable function H for $0 \leq t \leq 10$, where time t is measured in minutes and temperature $H(t)$ is measured in degrees Celsius. Values of $H(t)$ at selected values of time t are shown in the table above.

(a) Use the data in the table to approximate the rate at which the temperature of the tea is changing at time $t = 3.5$. Show the computations that lead to your answer.

(b) Using correct units, explain the meaning of $\frac{1}{10} \int_0^{10} H(t) dt$ in the context of this problem. Use a

trapezoidal sum with the four subintervals indicated by the table to estimate $\frac{1}{10} \int_0^{10} H(t) dt$.

(c) Evaluate $\int_0^{10} H'(t) dt$. Using correct units, explain the meaning of the expression in the context of this problem.

(d) At time $t = 0$, biscuits with temperature 100°C were removed from an oven. The temperature of the biscuits at time t is modeled by a differentiable function B for which it is known that $B'(t) = -13.84e^{-0.173t}$. Using the given models, at time $t = 10$, how much cooler are the biscuits than the tea?

Part A

1 point is earned for the answer

$$H'(3.5) \approx \frac{H(5) - H(2)}{5 - 2}$$

$$= \frac{52 - 60}{3} = -2.666 \text{ or } -2.667 \text{ degrees Celsius per minute}$$



0	1
---	---

The student response earns all of the following points:

1 point is earned for the answer

$$H'(3.5) \approx \frac{H(5) - H(2)}{5 - 2}$$

6-7b

$$= \frac{52 - 60}{3} = -2.666 \text{ or } -2.667 \text{ degrees Celsius per minute}$$

Part B

1 point is earned for meaning of the expression

1 point is earned for trapezoidal sum

1 point is earned for estimate

$\frac{1}{10} \int_0^{10} H(t) dt$ is the average temperature of the tea, in degrees Celsius, over the 10 minutes.

$$\begin{aligned} \frac{1}{10} \int_0^{10} H(t) dt &\approx \frac{1}{10} \left(2 \cdot \frac{66+60}{2} + 3 \cdot \frac{60+52}{2} + 4 \cdot \frac{52+44}{2} + 1 \cdot \frac{44+33}{2} \right) \\ &= 52.95 \end{aligned}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for meaning of the expression

1 point is earned for trapezoidal sum

1 point is earned for estimate

$\frac{1}{10} \int_0^{10} H(t) dt$ is the average temperature of the tea, in degrees Celsius, over the 10 minutes.

$$\begin{aligned} \frac{1}{10} \int_0^{10} H(t) dt &\approx \frac{1}{10} \left(2 \cdot \frac{66+60}{2} + 3 \cdot \frac{60+52}{2} + 4 \cdot \frac{52+44}{2} + 1 \cdot \frac{44+33}{2} \right) \\ &= 52.95 \end{aligned}$$

Part C

6-7b

1 point is earned for the value of integral

1 point is earned for meaning of the expression

$$\int_0^{10} H'(t) dt = H(10) - H(0) = 43 - 66 = -23$$

The temperature of the tea drops 23 degrees Celsius from time $t = 0$ to time $t = 10$ minutes.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the value of integral

1 point is earned for meaning of the expression

$$\int_0^{10} H'(t) dt = H(10) - H(0) = 43 - 66 = -23$$

The temperature of the tea drops 23 degrees Celsius from time $t = 0$ to time $t = 10$ minutes.

Part D

1 point is earned for integrand

1 point is earned for using $B(0) = 100$

1 point is earned for answer $B(10) = 100 + \int_0^{10} B'(t) dt = 34.18275$; $H(10) - H(0) = 8.817$

The biscuits are 8.817 degrees Celsius cooler than the tea.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for integrand

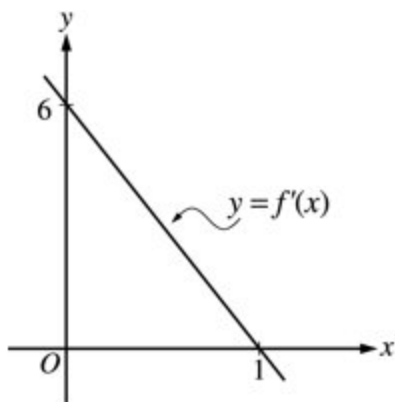
6-7b

1 point is earned for using $B(0) = 100$

1 point is earned for answer

$$B(10) = 100 + \int_0^{10} B'(t) dt = 34.18275; H(10) - H(0) = 8.817$$

The biscuits are 8.817 degrees Celsius cooler than the tea.

26.

The graph of f' , the derivative of f , is the line shown in the figure above. If $f(0) = 5$, then $f(1) =$

- (A) 0
- (B) 3
- (C) 6
- (D) 8
- (E) 11



27. $\int_0^1 \sqrt{x}(x+1)dx =$

- (A) 0
- (B) 1
- (C) $\frac{16}{15}$
- (D) $\frac{7}{5}$
- (E) 2



6-7b

28.

What are all values of k for which $\int_{-3}^k x^2 dx = 0$?

(A) -3



(B) 0

(C) 3

(D) -3 and 3

(E) -3, 0, 3

29.

$$\int_1^2 \frac{x-4}{x^2} dx$$

(A) $-\frac{1}{2}$ (B) $\ln 2 - 2$ (C) $\ln 2$

(D) 2

(E) $\ln 2 + 2$

6-7b

30.  The rate at which people enter an auditorium for a rock concert is modeled by the function R given by $R(t) = 1380t^2 - 675t^3$ for $0 \leq t \leq 2$ hours; $R(t)$ is measured in people per hour. No one is in the auditorium at time $t = 0$, when the doors open. The doors close and the concert begins at time $t = 2$.
- (a) How many people are in the auditorium when the concert begins?
- (b) Find the time when the rate at which people enter the auditorium is a maximum. Justify your answer.
- (c) The total wait time for all the people in the auditorium is found by adding the time each person waits, starting at the time the person enters the auditorium and ending when the concert begins. The function w models the total wait time for all the people who enter the auditorium before time t . The derivative of w is given by $w'(t) = (2 - t)R(t)$. Find $w(2) - w(1)$, the total wait time for those who enter the auditorium after time $t = 1$.
- (d) On average, how long does a person wait in the auditorium for the concert to begin? Consider all people who enter the auditorium after the doors open, and use the model for total wait time from part (c).

Part A

1 point is earned for integral

1 point is earned for answer

$$\int_0^2 R(t) dt = 980 \text{ people}$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for integral

1 point is earned for answer

$$\int_0^2 R(t) dt = 980 \text{ people}$$

Part B

1 point is earned if considers $R'(t) = 0$

1 point is earned for interior critical point

1 point is earned for the answer plus justification

6-7b

$R'(t) = 0$ when $t = 0$ and $t = 1.36296$

The maximum rate may occur at 0, $a = 1.36296$, or 2.

$$R(0) = 0$$

$$R(a) = 854.527$$

$$R(2) = 120$$

The maximum rate occurs when $t = 1.362$ or 1.363 .



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned if considers $R'(t) = 0$

1 point is earned for interior critical point

1 point is earned for the answer plus justification

$R'(t) = 0$ when $t = 0$ and $t = 1.36296$

The maximum rate may occur at 0, $a = 1.36296$, or 2.

$$R(0) = 0$$

$$R(a) = 854.527$$

$$R(2) = 120$$

The maximum rate occurs when $t = 1.362$ or 1.363 .

Part C

1 point is earned for integral

1 point is earned for answer

$$w(2) - w(1) = \int_1^2 w'(t) dt = \int_1^2 (2 - t)R(t) dt = 387.5$$

The total wait time for those who enter the auditorium after time $t = 1$ is 387.5 hours.

6-7b



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for integral

1 point is earned for answer

$$w(2) - w(1) = \int_1^2 w'(t) dt = \int_1^2 (2 - t)R(t) dt = 387.5$$

The total wait time for those who enter the auditorium after time $t = 1$ is 387.5 hours.

Part D

1 point is earned for integral

1 point is earned for answer

$$\frac{1}{980}w(2) = \frac{1}{980} \int_0^2 (2 - t)R(t) dt = 0.77551$$

On average, a person waits 0.775 or 0.776 hour.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for integral

1 point is earned for answer

$$\frac{1}{980}w(2) = \frac{1}{980} \int_0^2 (2 - t)R(t) dt = 0.77551$$

On average, a person waits 0.775 or 0.776 hour.

6-7b

31.

x	-4	-3	-2	-1
$f(x)$	0.75	-1.5	-2.25	-1.5
$f'(x)$	-3	-1.5	0	1.5

The table above gives values of a function f and its derivative at selected values of x . If f' is continuous on the interval $[-4, -1]$ what is the value of $\int_{-4}^{-1} f'(x) dx$?

(A) -4.5

(B) -2.25

(C) 0

(D) 2.25

(E) 4.5

32.

If n is a known positive integer, for what value of k is $\int_1^k x^{n-1} dx = \frac{1}{n}$?

(A) 0

(B) $\left(\frac{2}{n}\right)^{1/n}$ (C) $\left(\frac{2n-1}{n}\right)^{1/n}$ (D) $2^{1/n}$ (E) 2^n

33.

 If $f'(x) = \sin\left(\frac{\pi e^x}{2}\right)$ and $f(0) = 1$, then $f(2) =$

(A) -1.819

(B) -0.843

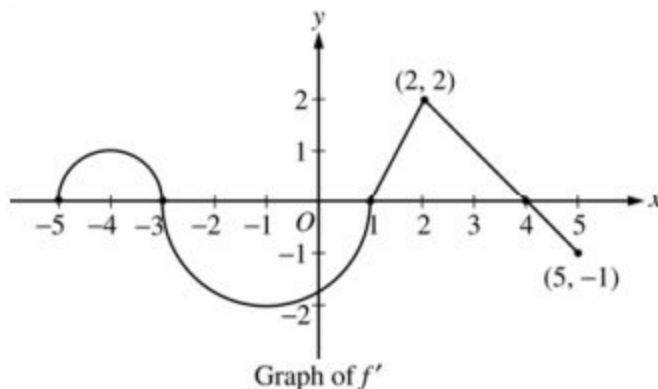
(C) -0.819

(D) 0.157

(E) 1.157

6-7b

34.



Let f be a function defined on the closed interval $-5 \leq x \leq 5$ with $f(1)=3$. The graph of f' the derivative of f , consists of two semicircles and two line segments, as shown above.

Find the absolute minimum value of $f(x)$ over the closed interval $-5 \leq x \leq 5$. Explain your reasoning.

Part D

1 point is earned if identifies $x=1$ as a candidate

1 point is earned if considers endpoint

1 point is earned for value and explanation

Candidates for the absolute minimum are where f' changes from negative to positive (at $x=1$) and at the endpoints ($x=-5, 5$).

$$f(-5) = 3 + \int_1^{-5} f'(x) dx = 3 - \frac{\pi}{2} + 2\pi > 3$$

$$f(1) = 3$$

$$f(5) = 3 + \int_1^5 f'(x) dx = 3 + \frac{3 \cdot 2}{2} - \frac{1}{2} > 3$$

The absolute minimum value of f on $[-5, 5]$ is $f(1)=3$.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned if identifies $x=1$ as a candidate

1 point is earned if considers endpoint

1 point is earned for value and explanation

6-7b

Candidates for the absolute minimum are where f' changes from negative to positive (at $x=1$) and at the endpoints ($x=-5,5$).

$$f(-5) = 3 + \int_1^{-5} f'(x) dx = 3 - \frac{\pi}{2} + 2\pi > 3$$

$$f(1) = 3$$

$$f(5) = 3 + \int_1^5 f'(x) dx = 3 + \frac{3 \cdot 2}{2} - \frac{1}{2} > 3$$

The absolute minimum value of f on $[-5, 5]$ is $f(1)=3$.

35. $\int_0^{\frac{\pi}{4}} \sin x \, dx =$

(A) $-\frac{\sqrt{2}}{2}$

(B) $\frac{\sqrt{2}}{2}$

(C) $-\frac{\sqrt{2}}{2} - 1$

(D) $-\frac{\sqrt{2}}{2} + 1$ ✓

(E) $\frac{\sqrt{2}}{2} - 1$

36. $\int_1^2 \frac{1}{x^2} dx =$

(A) $-\frac{1}{2}$

(B) $\frac{7}{24}$

(C) $\frac{1}{2}$ ✓

(D) 1

(E) $2\ln 2$

37. $\int_0^x \sin t \, dt =$

(A) $\sin x$

(B) $-\cos x$

(C) $\cos x$

(D) $\cos x - 1$

(E) $1 - \cos x$ ✓

6-7b

38.

If f is a continuous function and if $F'(x)=f(x)$ for all real numbers x , then $\int_1^3 f(2x)dx =$

- (A) $2F(3)-2F(1)$
- (B) $\frac{1}{2}F(3) - \frac{1}{2}F(1)$
- (C) $2F(6)-2F(2)$
- (D) $F(6)-F(2)$
- (E) $\frac{1}{2}F(6) - \frac{1}{2}F(2)$



39.



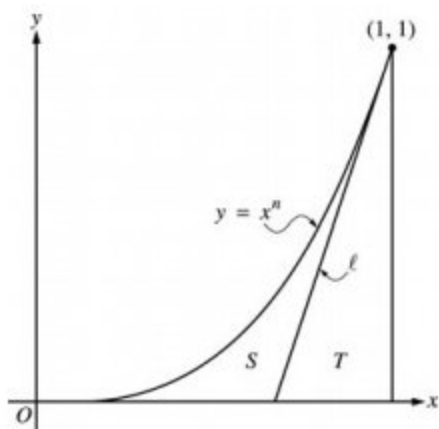
Let $F(x)$ be an antiderivative of $\frac{(\ln x)^3}{x}$. If $F(1)=0$, then $F(9)=$

- (A) 0.048
- (B) 0.144
- (C) 5.827
- (D) 23.308
- (E) 1,640.250



6-7b

40.



Let l be the line tangent to the graph of $y = x^n$ at the point $(1, 1)$, where $n > 1$, as shown above.

Find $\int_0^1 x^n dx$ in terms of n .

Part A

1 point is earned for the antiderivative of x^n

1 point is earned for the answer

$$\int_0^1 x^n dx = \frac{x^{n+1}}{n+1} \bigg|_0^1 = \frac{1}{n+1}$$

✓

0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the antiderivative of x^n

1 point is earned for the answer

$$\int_0^1 x^n dx = \frac{x^{n+1}}{n+1} \bigg|_0^1 = \frac{1}{n+1}$$

6-7b

41.
$$\int_0^1 (3x - 2)^2 dx =$$

(A) $-\frac{7}{3}$

(B) $-\frac{7}{9}$

(C) $\frac{1}{9}$

(D) 1

(E) 3



42.
$$\int_0^{\frac{\pi}{2}} \frac{\cos \theta}{\sqrt{1 + \sin \theta}} d\theta =$$

(A) $-2(\sqrt{2} - 1)$

(B) $-2\sqrt{2}$

(C) $2\sqrt{2}$

(D) $2(\sqrt{2} - 1)$

(E) $2(\sqrt{2} + 1)$



43. If the function f has a continuous derivative on $[0, c]$, then $\int_0^c f'(x) dx =$

(A) $f(c) - f(0)$

(B) $|f(c) - f(0)|$

(C) $f(c)$

(D) $f(x) + c$

(E) $f'(c) - f'(0)$



44. If $\int_0^k (2kx - x^2) dx = 18$, then $k =$

(A) -9

(B) -3

(C) 3

(D) 9

(E) 18



6-7b

45. $\int_1^e \frac{x^2 + 1}{x} dx =$

(A) $\frac{e^2 - 1}{2}$

(B) $\frac{e^2 + 1}{2}$ ✓

(C) $\frac{e^2 + 2}{2}$

(D) $\frac{e^2 - 1}{e^2}$

(E) $\frac{2e^2 - 8e + 6}{3e}$

6-7b

46. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (days)	0	5	10	15	20
$M(t)$ (number of 1-cent coins)	0	300	500	870	1100

A class collected 1-cent coins to raise money for classroom supplies. Students in the class deposited the coins into a container where all the coins were kept until the end of the collection period. For time t days, $0 \leq t \leq 20$, the total number of coins in the container is modeled by a twice-differentiable function M . Values of $M(t)$ at various times t are shown in the table above.

(a) Use a midpoint Riemann sum with the two subintervals indicated by the table to approximate

$\int_0^{20} M(t) \, dt$. Show the computations that lead to your answer.

(b) Find the value of $\int_0^5 M'(3t) \, dt$. Show the computations that lead to your answer.

(c) Using correct units, approximate $M'(15)$ using the average rate of change of M over the interval $10 \leq t \leq 20$.

(d) Explain the meaning of $M'(15)$ in the context of the problem.

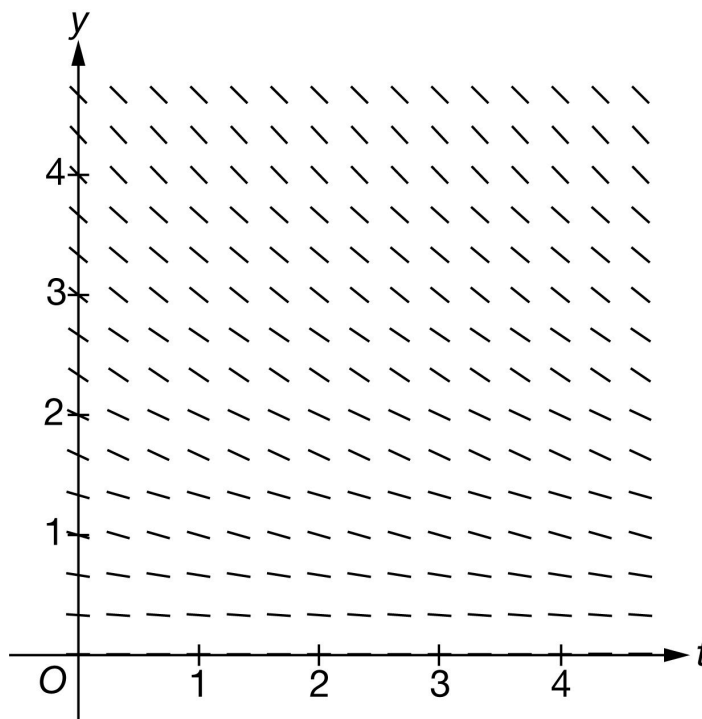
(e) It is known that $M'(10) = 55$. Write an equation for the line tangent to the graph of $y = M(t)$ at $t = 10$.

(f) It is known that $M''(t) > 0$ for $10 \leq t \leq 15$. If $M(14)$ is approximated using the equation from part (e), would the approximation be an overestimate or underestimate for the actual value of $M(14)$? Give a reason for your answer.

(g) For $0 \leq t \leq 20$, the total number of coins collected can also be modeled by the differentiable function R , where R is a function of t . Assume the weight of each coin is 3.5 grams. Write an expression for the rate of change, in grams per day, of the total weight of the coins at $t = 11$ days.

6-7b

(h) Explain why the following slope field cannot represent the differential equation $\frac{dy}{dt} = 0.3y$.



Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point (Form of midpoint sum): Philosophy: The response should provide clear evidence adding the area of two rectangles whose heights are found using the midpoint of the interval.

Acceptable evidence:

$$\cdot 10 \cdot (M(5) + M(15))$$

$$\cdot 10 \cdot M(5) + 10 \cdot M(15)$$

$$\cdot (10 - 0) \cdot 300 + (20 - 10) \cdot 870$$

$$\cdot 10 \cdot (300 + 870) \text{ or } 10 \cdot 300 + 10 \cdot 870$$

To earn this point the response can have at most one mistake in the expression or equation. So, expressions of the form $\frac{10}{b} \cdot (M(5) + M(15))$, for any nonzero value of b would earn the point. Note: The value $\frac{10}{b}$ might be distributed across the sum.

Expressions of the form $\frac{a}{b} \cdot (M(5) + M(15))$ for $a \neq 10$ will not earn this point.

Once the first point is earned, it cannot be lost.

Second Point (Answer)

6-7b

To earn the point, the answer must be correct. It does not need to be simplified, but if it is, the simplification must be correct.

The answer point cannot be earned unless the first (Form of midpoint sum) point was earned.

Minimal complete answers that earn both points:

$$\cdot 10 \cdot (300 + 870)$$

$$\cdot 10 \cdot 300 + 10 \cdot 870$$



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Form of midpoint sum
- ☐ Answer

Solution:

$$\begin{aligned} \int_0^{20} M(t) dt &\approx M(5)(10 - 0) + M(15)(20 - 10) \\ &= 300 \cdot 10 + 870 \cdot 10 = 11,700 \end{aligned}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point (Substitution) Philosophy: This point is earned when the response has all three components of the substitution done correctly.

Uses $u = 3t$, perhaps implicitly (any variable could be used for u)

Uses the correct constant of $\frac{1}{3}$

Uses the correct limits for their substitution.

· The limits might not be seen until the final evaluation, and at that point, response may have returned to the variable t .

· Limits for t attached to an integral involving u or vice versa will not earn this substitution point but could still earn the answer point, if the numerical answer is correct.

6-7b

Second point: (Fundamental Theorem of Calculus) Philosophy: This point is earned when the FTC has been applied correctly to either $\int M'(3t) dt$ or $\int M'(u) du$.

The point is earned for any of the following:

- The equation $\int M'(3t) dt = M(3t)$ or $\int M'(u) du = M(u)$
- $\frac{1}{3}M(3t)|_0^5$ or $\frac{1}{3}M(u)|_0^{15}$
- $M(15) - M(0)$, $M(15)$, or $M(0)$

This point can be earned if the substitution point is not.

Note: $\int_a^b M'(u) du = M'(b) - M'(a)$ does not earn the point. The FTC point is for the response communicating that the antiderivative of M' is M .

Third Point (Answer)

Must be the correct numerical answer.

Must have earned the second point for the FTC.

Might have not earned the first point for substitution because of an incorrect use of limits, but will still earn the third point for the correct answer.

$M(15)$ and $M(0)$ must be evaluated; the expression $\frac{1}{3}(M(15) - M(0))$ by itself does not earn the third point.

The answer may be unsimplified, but if simplified it must be done correctly.

Scoring common responses:

The minimal work required to earn all three points is $\frac{1}{3}(870 - 0)$ or $\frac{1}{3} \cdot 870$

The following responses would earn the first two points and be eligible for the answer point if the response continues:

- $\frac{1}{3}M(3t)|_0^5$
- $\frac{1}{3}(M(15) - M(0))$

Responses of only the following would earn only the second point:

- $M(15) - M(0)$
- $M(5) - M(0)$

6-7b

$$\cdot M(15)$$

$$\cdot \int_0^5 M'(u) \, du = M(5)$$

$$\cdot M(3t)|_0^5 \text{ or } M(u)|_0^{15}$$

The following responses would earn no points:

· 290 in the absence of one of the first two points.

$$\cdot \int_0^5 M'(3t) \, dt = M(5)$$



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

Let $u = 3t$. Then $du = 3 \, dt$.

When $t = 0$, $u = 0$.

When $t = 5$, $u = 15$.

$$\int_0^5 M'(3t) \, dt = \frac{1}{3} \int_0^{15} M'(u) \, du$$

$$= \frac{1}{3}(M(15) - M(0))$$

$$= \frac{1}{3}(870 - 0) = 290$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point (Approximation)

The response must be a difference and a quotient.

6-7b

The answer does not need to be simplified, but if it is simplified it must be done correctly.

The minimal work that must be shown to earn this point is either $\frac{1100-500}{10}$ or $\frac{600}{20-10}$.

Second point (Units)

The units must be attached to a correct or incorrect value.

The response will not be penalized for any verbal attempt to discuss the answer after the correct approximation with units is presented.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ Approximation

☐ Units

Solution:

$$M'(15) \approx \frac{M(20) - M(10)}{20 - 10} = \frac{1100 - 500}{10} = 60 \text{ coins per day}$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The interpretation must include three components:

- indicate the rate of change
- the specific context of the number of coins in the container
- the specific point in time, $t = 15$

Responses do not need to include correct units or any units.

The word “increasing” may be used in place of the word “changing”.

The response is not earned for “the rate of coins deposited on day 15 “ as this does not indicate a change in the rate of deposit.

“the rate of change of M “ without specifying that M is the number of coins deposited, does not earn the point.

Acceptable ways to provide the specific point in time:

6-7b

- On day 15 or after 15 days.
- At $t = 15$ (missing the unit of time)
- On week 15 (incorrect unit of time)

A response that clearly indicates an interval of time (e.g. “during the first 15 days” or “from day 15 on”) does not earn the point.



0	1
---	---

The student response accurately includes the criteria below.

☐ Explanation

Solution:

$M'(15)$ is the rate at which the number of 1-cent coins in the container is changing, in coins per day, on day 15.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The use of $M(t)$ in place of y earns the point.

The use of x in place of t earns the point.

Tangent lines do not need to be simplified, but any simplification must be done correctly.

Any correct form of the tangent line earns the point. For example:

- $y = 500 + 55(t - 10)$
- $y - 500 = 55(t - 10)$
- $y = 55t - 550 + 500$
- $y = 55t - 50$



0	1
---	---

The student response accurately includes the criteria below.

6-7b

☐ Tangent line equation
Solution:

$$y = M'(10)(t - 10) + M(10)$$

$$y = 55(t - 10) + 500$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the point the response must include $M''(t) > 0$ and underestimate.

“ $M(t)$ is concave up” can be used instead of $M''(t) > 0$.

Information about $M'(t)$ by itself without reference to $M''(t) > 0$ is not sufficient and does not earn this point.

“Underestimate because it is concave up” is not sufficient by itself because of the use of “it”. The response must clearly, directly or indirectly, indicate to what “it” is referring.

If a response makes the correct conclusion and states, $M''(t) > 0$, and then goes on and makes an incorrect statement or a statement that cannot be verified will not earn this point.

For example: “ $M'(t)$ is concave up, underestimate” does not earn this point.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer with reason
Solution:

Because $M''(t) > 0$, the tangent line approximation would produce an underestimate.

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The correct expression of only $3.5R'(11)$ earns this point.

Only expression $3.5M'(11)$ also earns this point.

No justifications or explanations are required.

6-7b

Any incorrect statement, for example $R'(11) = 3.5M'(11)$, does not earn this point.



0	1
---	---

The student response accurately includes the criteria below.

☐ Expression

Solution:

The rate of change, in grams per day, of the total weight of the coins at time $t = 11$ days is $3.5R'(11)$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Philosophy: Response clearly references what part of the differential equation is different from the slope field and the response needs only to clearly and explicitly reference one of them.

The response does not need to explicitly reference both the differential equation and the given slope field.

The distinction can be made with explicit reference to one of the two things and an implicit reference to the other

For example:

· “ $\frac{dy}{dt}$ is positive, not negative”. We will infer that negative is referring to the slopes shown in the slope field.

· “This would be correct for a negative derivative, but $\frac{dy}{dt}$ is positive.” We will read “This” as “the slope field” in this sentence.

· “All of the slopes in the slope field are negative, not positive.” We will infer that the values of $\frac{dy}{dt}$ at points in the first quadrant are positive.

A response that lacks even an implicit reference to either the differential equation or the slope field will not earn the point. For example:

· “ $\frac{dy}{dt}$ is positive” lacks any reference to the slope field and so will not earn the point.

· “All the slopes in the graph are negative” lacks any reference to the differential equation, and so will not earn the point.

· Vague answers such as “The slope field has negative points” or “The field is decreasing” will not earn the point.

6-7b



0

1

The student response accurately includes the criteria below.

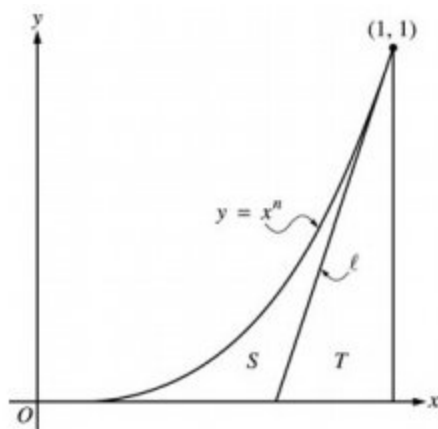
☐ Explanation

Solution:

When $y > 0$, $\frac{dy}{dt} > 0$. However, in the given slope field, the slopes of the line segments for $y > 0$ are negative.

6-7b

47.



Let l be the line tangent to the graph of $y = x^n$ at the point $(1, 1)$, where $n > 1$, as shown above.

(a) Find $\int_0^1 x^n dx$ in terms of n .

(b) Let T be the triangular region bounded by l , the x -axis, and the line $x = 1$. Show that the area of T is $1/2n$.

(c) Let S be the region bounded by the graph of $y = x^n$, the line l , and the x -axis. Express the area of S in terms of n and determine the value of n that maximizes the area of S .

Part A

1 point is earned for the antiderivative of x^n

1 point is earned for the answer

$$\int_0^1 x^n dx = \frac{x^{n+1}}{n+1} \Big|_0^1 = \frac{1}{n+1}$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the antiderivative of x^n

1 point is earned for the answer

$$\int_0^1 x^n dx = \frac{x^{n+1}}{n+1} \Big|_0^1 = \frac{1}{n+1}$$

6-7b

Part B

1 point is earned if slope of line l is n

1 point is earned if base of T is $\frac{1}{n}$

1 point is earned if shows area is $\frac{1}{2n}$

Let b be the length of the base of triangle T .

$\frac{1}{b}$ is the slope of line l , which is n

$$\text{Area}(T) = \frac{1}{2}b(1) = \frac{1}{2n}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned if slope of line l is n

1 point is earned if base of T is $\frac{1}{n}$

1 point is earned if shows area is $\frac{1}{2n}$

Let b be the length of the base of triangle T .

$\frac{1}{b}$ is the slope of line l , which is n

$$\text{Area}(T) = \frac{1}{2}b(1) = \frac{1}{2n}$$

Part C

6-7b

1 point is earned for the area of S in terms of n

1 point is earned for the derivative

1 point is earned if sets derivative equal to 0

1 point is earned if solves for n

$$\begin{aligned}\text{Area}(S) &= \int_0^1 x^n dx - \text{Area}(T) \\ &= \frac{1}{n+1} - \frac{1}{2n} \\ \frac{d}{dn}\text{Area}(S) &= -\frac{1}{(n+1)^2} + \frac{1}{2n^2} = 0 \\ 2n^2 &= (n+1)^2 \\ \sqrt{2}n &= (n+1) \\ n &= \frac{1}{\sqrt{2}-1} = 1 + \sqrt{2}\end{aligned}$$



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned for the area of S in terms of n

1 point is earned for the derivative

1 point is earned if sets derivative equal to 0

1 point is earned if solves for n

$$\begin{aligned}\text{Area}(S) &= \int_0^1 x^n dx - \text{Area}(T) \\ &= \frac{1}{n+1} - \frac{1}{2n} \\ \frac{d}{dn}\text{Area}(S) &= -\frac{1}{(n+1)^2} + \frac{1}{2n^2} = 0 \\ 2n^2 &= (n+1)^2 \\ \sqrt{2}n &= (n+1) \\ n &= \frac{1}{\sqrt{2}-1} = 1 + \sqrt{2}\end{aligned}$$

6-7b

48. Let R be the region enclosed by the graphs of $y = e^x$, $y = (x - 1)^2$, and the line $x = 1$.
- (a) Find the area of R .
- (b) Find the volume of the solid generated when R is revolved about the x -axis.
- (c) Set up, but do not integrate, an integral expression in terms of a single variable for the volume of the solid generated when R is revolved about the y -axis.

Part A

1 point is earned for definite integral

1 point is **deducted** for incorrect limits

1 point is **deducted** for reversal

1 point is earned for anti derivative

1 point is earned for evaluation

$$\begin{aligned}
 A &= \int_0^1 e^x - (x - 1)^2 dx \\
 &= \int_0^1 e^x - x^2 + 2x - 1 dx \\
 &= e^x \Big|_0^1 - \frac{1}{3}(x - 1)^3 \Big|_0^1 \\
 &= (e - 1) - \frac{1}{3} = e - \frac{4}{3}
 \end{aligned}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for definite integral

1 point is **deducted** for incorrect limits

1 point is **deducted** for reversal

1 point is earned for anti derivative

1 point is earned for evaluation

6-7b

$$\begin{aligned}
 A &= \int_0^1 e^x - (x-1)^2 dx \\
 &= \int_0^1 e^x - x^2 + 2x - 1 dx \\
 &= e^x \Big|_0^1 - \frac{1}{3}(x-1)^3 \Big|_0^1 \\
 &= (e-1) - \frac{1}{3} = e - \frac{4}{3}
 \end{aligned}$$

Part B

2 points are earned for definite integral

2 points are **deducted** for faulty f, g, or u

1 point is earned for correct antiderivative

1 point is earned for evaluation of anti derivative

$$\begin{aligned}
 V &= \pi \int_0^1 e^{2x} - (x-1)^4 dx \\
 &= \pi \left[\frac{e^{2x}}{2} \right] \Big|_0^1 - \pi \left[\frac{1}{5}(x-1)^5 \right] \Big|_0^1 \\
 &= \pi \left[\left(\frac{e^2}{2} - \frac{1}{2} \right) - \frac{1}{5} \right] = \pi \left(\frac{e^2}{2} - \frac{7}{10} \right)
 \end{aligned}$$

or

$$\begin{aligned}
 V &= 2\pi \int_0^1 y[1 - (1 - \sqrt{y})] dy + 2\pi \int_1^e y(1 - \ln y) dy \\
 &= 2\pi \cdot \frac{2}{5} y^{5/2} \Big|_0^1 + 2\pi \left[\frac{1}{2} y^2 - \left(\frac{1}{2} y^2 \ln y - \frac{1}{4} y^2 \right) \right] \Big|_1^e \\
 &= \frac{4}{5} \pi + 2\pi \left[\frac{1}{4} e^2 - \frac{3}{4} \right] = \pi \left(\frac{e^2}{2} - \frac{7}{10} \right)
 \end{aligned}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 points are earned for definite integral

6-7b

2 points are **deducted** for faulty f, g, or u

1 point is earned for correct antiderivative

1 point is earned for evaluation of anti derivative

$$\begin{aligned} V &= \pi \int_0^1 e^{2x} - (x-1)^4 dx \\ &= \pi \left[\frac{e^{2x}}{2} \right]_0^1 - \pi \left[\frac{1}{5} (x-1)^5 \right]_0^1 \\ &= \pi \left[\left(\frac{e^2}{2} - \frac{1}{2} \right) - \frac{1}{5} \right] = \pi \left(\frac{e^2}{2} - \frac{7}{10} \right) \end{aligned}$$

or

$$\begin{aligned} V &= 2\pi \int_0^1 y [1 - (1 - \sqrt{y})] dy + 2\pi \int_1^e y (1 - \ln y) dy \\ &= 2\pi \cdot \frac{2}{5} y^{5/2} \Big|_0^1 + 2\pi \left[\frac{1}{2} y^2 - \left(\frac{1}{2} y^2 \ln y - \frac{1}{4} y^2 \right) \right]_1^e \\ &= \frac{4}{5} \pi + 2\pi \left[\frac{1}{4} e^2 - \frac{3}{4} \right] = \pi \left(\frac{e^2}{2} - \frac{7}{10} \right) \end{aligned}$$

Part C

2 points are earned for Definite integral

2 points are **deducted** for f, g, or u error

1 point is **deducted** for limit error, K error and integrand reversal

$$V = 2\pi \int_0^1 x [e^x - (x-1)^2] dx$$

or

$$V = \pi \int_0^1 1 - (1 - \sqrt{y})^2 dy + \pi \int_1^e 1 - (\ln y)^2 dy$$



0	1	2
---	---	---

6-7b

The student response earns all of the following points:

2 points are earned for Definite integral

2 points are **deducted** for f, g, or u error

1 point is **deducted** for limit error, K error and integrand reversal

$$V = 2\pi \int_0^1 x \left[e^x - (x-1)^2 \right] dx$$

or

$$V = \pi \int_0^1 1 - (1 - \sqrt{y})^2 dy + \pi \int_1^e 1 - (\ln y)^2 dy$$

6-7b

49.  Let R be the region in the first quadrant under the graph of $y = \frac{1}{\sqrt{x}}$ for $4 \leq x \leq 9$.

- (a) Find the area of R .
- (b) If the line $x = k$ divides the region R into two regions of equal area, what is the value of k ?
- (c) Find the volume of the solid whose base is the region R and whose cross sections cut by planes perpendicular to the x -axis are squares.

Part A

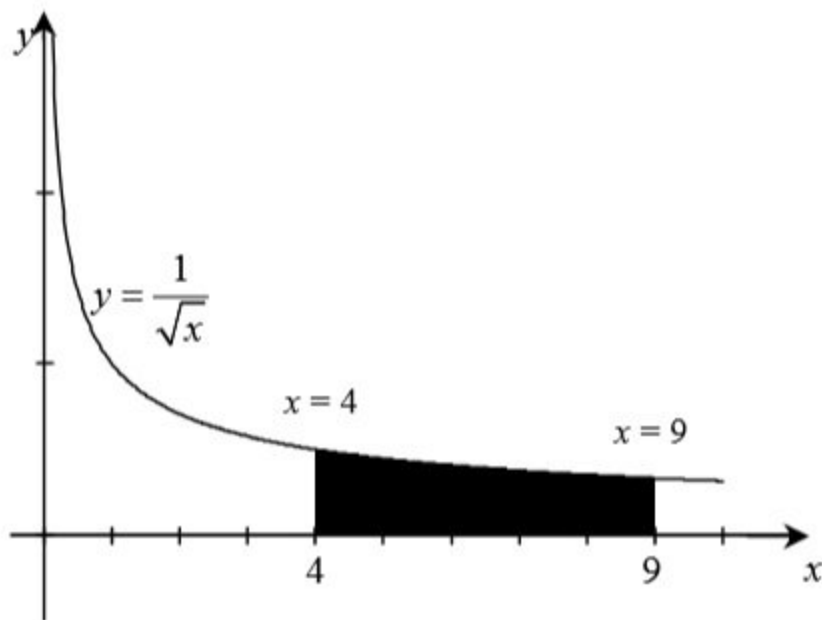
1 point is earned for the correct limits in x

1 point is earned for the correct integrand including constant must be $\frac{1}{\sqrt{x}}$

1 point is earned for the correct answer

$$\int_4^9 \frac{dx}{\sqrt{x}} = 2$$

0/1 if the integrand is incorrect, wrong limits are read



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the correct limits in x

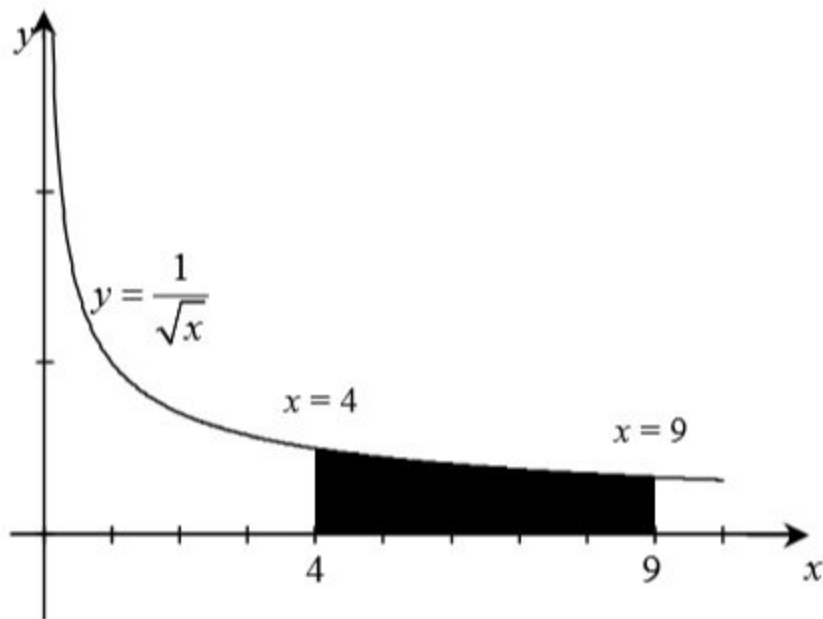
6-7b

1 point is earned for the correct integrand including constant must be $\frac{1}{\sqrt{x}}$

1 point is earned for the correct answer

$$\int_4^9 \frac{dx}{\sqrt{x}} = 2$$

0/1 if the integrand is incorrect, wrong limits are read

**Part B**

1 point is earned for the correct demonstration of variable limit using k as upper or lower limit with 4 or 9 (or imported limit)

$$\int_4^k \frac{dx}{\sqrt{x}} \text{ or } \int_k^9 \frac{dx}{\sqrt{x}}$$

1 point is earned for the correct equation involving the two halves of R that when solved will give answer with demonstrated algorithm that knows when to stop

1 point is earned for the correct answer

0/1 if answer from equation not involving relevant areas

1/3 bald answer

6-7b

$$\int_4^k \frac{dx}{\sqrt{x}} = 1$$

$$2\sqrt{x}]_4^k = 1$$

$$2\sqrt{k} - 2\sqrt{4} = 1$$

$$k = \frac{25}{4}$$

$$\left(\text{or } \int_k^9 \frac{dx}{\sqrt{x}} = 1 \text{ or } \int_4^k \frac{dx}{\sqrt{x}} = \int_k^9 \frac{dx}{\sqrt{x}} \right)$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the correct demonstration of variable limit using k as upper or lower limit with 4 or 9 (or imported limit)

$$\int_4^k \frac{dx}{\sqrt{x}} \text{ or } \int_k^9 \frac{dx}{\sqrt{x}}$$

1 point is earned for the correct equation involving the two halves of R that when solved will give answer with demonstrated algorithm that knows when to stop

1 point is earned for the correct answer

0/1 if answer from equation not involving relevant areas

1/3 bald answer

$$\int_4^k \frac{dx}{\sqrt{x}} = 1$$

$$2\sqrt{x}]_4^k = 1$$

$$2\sqrt{k} - 2\sqrt{4} = 1$$

$$k = \frac{25}{4}$$

$$\left(\text{or } \int_k^9 \frac{dx}{\sqrt{x}} = 1 \text{ or } \int_4^k \frac{dx}{\sqrt{x}} = \int_k^9 \frac{dx}{\sqrt{x}} \right)$$

Part C

1 point is earned for the correct indefinite limits using x

1 point is earned for the correct integrand including constant

6-7b

1 point is earned for the correct answer

0/1 if integrand is incorrect

$$\begin{aligned}\text{Volume} &= \int_4^9 \left(\frac{1}{\sqrt{x}} \right)^2 dx \\ &= \int_4^9 \frac{dx}{x} = \ln x \Big|_4^9 = \ln \frac{9}{4}\end{aligned}$$

(or 0.811)



0	1	2	3
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The student response earns all of the following points:

1 point is earned for the correct indefinite limits using x

1 point is earned for the correct integrand including constant

1 point is earned for the correct answer

0/1 if integrand is incorrect

$$\begin{aligned}\text{Volume} &= \int_4^9 \left(\frac{1}{\sqrt{x}} \right)^2 dx \\ &= \int_4^9 \frac{dx}{x} = \ln x \Big|_4^9 = \ln \frac{9}{4}\end{aligned}$$

(or 0.811)

6-7b

50.

x	2	3	5	8	13
$f(x)$	1	4	-2	3	6

Let f be a function that is twice differentiable for all real numbers. The table above gives values of f for selected points in the closed interval $2 \leq x \leq 13$.

(a) Estimate $f'(4)$. Show the work that leads to your answer.

(b) Evaluate $\int_2^{13} (3 - 5f'(x)) dx$. Show the work that leads to your answer.

(c) Use a left Riemann sum with subintervals indicated by the data in the table to approximate

$\int_2^{13} f(x) dx$. Show the work that leads to your answer.

(d) Suppose $f(5) = 3$ and $f''(x) < 0$ for all x in the closed interval $5 \leq x \leq 8$. Use the line tangent to the graph of f at $x = 5$ to show that $f(7) \leq 4$. Use the secant line for the graph of f on $5 \leq x \leq 8$ to show that $f(7) \geq 4/3$.

Part A

1 point is earned for the answer

$$f'(4) \approx \frac{f(5) - f(3)}{5 - 3} = -3$$



0	1
---	---

The student response earns all of the following points:

1 point is earned for the answer

$$f'(4) \approx \frac{f(5) - f(3)}{5 - 3} = -3$$

Part B

1 point is earned if uses Fundamental Theorem of Calculus

1 point is earned for the answer

6-7b

$$\begin{aligned}\int_2^{13} (3 - 5f'(x)) dx &= \int_2^{13} 3dx - 5 \int_2^{13} f'(x) dx \\ &= 3(13 - 2) - 5(f(13) - f(2)) = 8\end{aligned}$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned if uses Fundamental Theorem of Calculus

1 point is earned for the answer

$$\begin{aligned}\int_2^{13} (3 - 5f'(x)) dx &= \int_2^{13} 3dx - 5 \int_2^{13} f'(x) dx \\ &= 3(13 - 2) - 5(f(13) - f(2)) = 8\end{aligned}$$

Part C

1 point is earned for left Riemann sum

1 point is earned for the answer

$$\int_2^{13} f(x) dx \approx f(2)(3 - 2) + f(3)(5 - 3) + f(5)(8 - 5) + f(8)(13 - 8) = 18$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for left Riemann sum

1 point is earned for the answer

$$\int_2^{13} f(x) dx \approx f(2)(3 - 2) + f(3)(5 - 3) + f(5)(8 - 5) + f(8)(13 - 8) = 18$$

6-7b

Part D

1 point is earned for the tangent line

1 point is earned for showing $f(7) \leq 4$

1 point is earned for secant line

1 point is earned if shows $f(7) \geq 4/3$

An equation for the tangent line is $y = -2 + 3(x - 5)$.

Since $f''(x) < 0$ for all x in the interval $5 \leq x \leq 8$, the line tangent to the graph of $y = f(x)$ at $x = 5$ lies above the graph for all x in the interval $5 < x \leq 8$.

Therefore, $f(7) \leq -2 + 3 \cdot 2 = 4$.

An equation for the secant line is $y = -2 + \frac{5}{3}(x - 5)$.

Since $f''(x) < 0$ for all x in the interval $5 \leq x \leq 8$, the secant line connecting $(5, f(5))$ and $(8, f(8))$ lies below the graph of $y = f(x)$ for all x in the interval $5 < x < 8$.

Therefore, $f(7) \geq -2 + \frac{5}{3} \cdot 2 = \frac{4}{3}$.



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned for the tangent line

1 point is earned for showing $f(7) \leq 4$

1 point is earned for secant line

1 point is earned if shows $f(7) \geq 4/3$

An equation for the tangent line is $y = -2 + 3(x - 5)$.

Since $f''(x) < 0$ for all x in the interval $5 \leq x \leq 8$, the line tangent to the graph of $y = f(x)$ at $x = 5$ lies above the graph for all x in the interval $5 < x \leq 8$.

6-7b

Therefore, $f(7) \leq -2 + 3 \cdot 2 = 4$.

An equation for the secant line is $y = -2 + \frac{5}{3}(x - 5)$.

Since $f''(x) < 0$ for all x in the interval $5 \leq x \leq 8$, the secant line connecting $(5, f(5))$ and $(8, f(8))$ lies below the graph of $y = f(x)$ for all x in the interval $5 < x < 8$.

Therefore, $f(7) \geq -2 + \frac{5}{3} \cdot 2 = \frac{4}{3}$.

6-7b

51. Let $F(x) = \int_0^x \sin(t^2) dt$ for $0 \leq x \leq 3$.

(a) Use the trapezoidal rule with four equal subdivisions of the closed interval $[0,1]$ to approximate $F(1)$.

(b) On what intervals is F increasing?

(c) If the average rate of change of F on the closed interval $[1,3]$ is k , find $\int_1^3 \sin(t^2) dt$ in terms of k .

Part A

1 point is earned for correctly using $\int_0^1 \sin(t^2) dt$ or

$$x_0 = 0, x_1 = \frac{1}{4}, x_2 = \frac{1}{2}, x_3 = \frac{3}{4}, x_4 = 1$$

$$F(1) = \int_0^1 \sin(t^2) dt$$

1 point is earned for correctly substituting $x = 0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1$ into $\sin(x^2)$

$$\approx \frac{(1-0)}{4} \cdot \frac{1}{2} \cdot [\sin 0^2 + 2 \sin \left(\frac{1}{4}\right)^2 + 2 \sin \left(\frac{1}{2}\right)^2 + 2 \sin \left(\frac{3}{4}\right)^2 + \sin 1^2]$$

1 point is earned for the correct answer using $\sin(x^2)$ -values in trapezoidal rule formula on $[0,1]$

$$\approx 0.316$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for correctly using $\int_0^1 \sin(t^2) dt$ or

$$x_0 = 0, x_1 = \frac{1}{4}, x_2 = \frac{1}{2}, x_3 = \frac{3}{4}, x_4 = 1$$

$$F(1) = \int_0^1 \sin(t^2) dt$$

1 point is earned for correctly substituting $x = 0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1$ into $\sin(x^2)$

6-7b

$$\approx \frac{(1-0)}{4} \cdot \frac{1}{2} \cdot [\sin 0^2 + 2 \sin \left(\frac{1}{4}\right)^2 + 2 \sin \left(\frac{1}{2}\right)^2 + 2 \sin \left(\frac{3}{4}\right)^2 + \sin 1^2]$$

1 point is earned for the correct answer using $\sin(x^2)$ -values in trapezoidal rule formula on $[0,1]$

$$\approx 0.316$$

Part B

1 point is earned for correctly using $F'(x) = \sin(x^2)$

$$F'(x) = \sin(x^2)$$

$$F'(x) = 0 \text{ when } x^2 = 0, \pi, 2\pi, \dots$$

1 point is earned for correctly finding zeros of F' (and eliminating extraneous zeros)

$$x = 0, \sqrt{\pi}, \sqrt{2\pi}$$

1 point is earned for correctly analyzing sign of F' within $[0,3]$



1 point is earned for correctly indicating F increases where student's F' is positive

F is increasing on $[0, \sqrt{\pi}]$ and on $[\sqrt{2\pi}, 3]$



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned for correctly using $F'(x) = \sin(x^2)$

$$F'(x) = \sin(x^2)$$

$$F'(x) = 0 \text{ when } x^2 = 0, \pi, 2\pi, \dots$$

1 point is earned for correctly finding zeros of F' (and eliminating extraneous zeros)

$$x = 0, \sqrt{\pi}, \sqrt{2\pi}$$

6-7b

1 point is earned for correctly analyzing sign of F' within $[0,3]$



1 point is earned for correctly indicating F increases where student's F' is positive

F is increasing on $[0, \sqrt{\pi}]$ and on $[\sqrt{2\pi}, 3]$

Part C

1 point is earned for correctly solving $k = \frac{F(3)-F(1)}{2}$

$$k = \frac{F(3)-F(1)}{2} = \frac{\int_1^3 \sin(t^2) dt}{2}$$

1 point is earned for the correct answer

$$\int_1^3 \sin(t^2) dt = 2k$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for correctly solving $k = \frac{F(3)-F(1)}{2}$

$$k = \frac{F(3)-F(1)}{2} = \frac{\int_1^3 \sin(t^2) dt}{2}$$

1 point is earned for the correct answer

$$\int_1^3 \sin(t^2) dt = 2k$$

6-7b

52.  At a certain height, a tree trunk has a circular cross section. The radius $R(t)$ of that cross section grows at a rate modeled by the function

$$\frac{dR}{dt} = \frac{1}{16}(3 + \sin(t^2)) \text{ centimeters per year}$$

for $0 \leq t \leq 3$, where time t is measured in years. At time $t = 0$, the radius is 6 centimeters. The area of the cross section at time t is denoted by $A(t)$.

(a) Write an expression, involving an integral, for the radius $R(t)$ for $0 \leq t \leq 3$. Use your expression to find $R(3)$.

(b) Find the rate at which the cross-sectional area $A(t)$ is increasing at time $t = 3$ years. Indicate units of measure.

(c) Evaluate $\int_0^3 A'(t) dt$. Using appropriate units, interpret the meaning of that integral in terms of cross-sectional area.

Part A

1 point is earned for the integral

1 point is earned for the expression for $R(t)$

1 point is earned for $R(3)$

$$R(t) = 6 + \int_0^t \frac{1}{16}(3 + \sin(x^2)) dx$$

$$R(3) = 6.610 \text{ or } 6.611$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the integral

1 point is earned for the expression for $R(t)$

1 point is earned for $R(3)$

$$R(t) = 6 + \int_0^t \frac{1}{16}(3 + \sin(x^2)) dx$$

$$R(3) = 6.610 \text{ or } 6.611$$

Part B

6-7b

1 point is earned for expression of $A(t)$

1 point is earned for expression of $A'(t)$

1 point is earned for answering with units

$$A(t) = \pi(R(t))^2$$

$$A'(t) = 2\pi R(t)R'(t)$$

$$A'(3) = 8.858 \text{ cm}^2/\text{year}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for expression of $A(t)$

1 point is earned for expression of $A'(t)$

1 point is earned for answering with units

$$A(t) = \pi(R(t))^2$$

$$A'(t) = 2\pi R(t)R'(t)$$

$$A'(3) = 8.858 \text{ cm}^2/\text{year}$$

Part C

1 point is earned for using Fundamental Theorem of Calculus

1 point is earned for value of $\int_0^3 A'(t)dt$

1 point is earned for meaning of $\int_0^3 A'(t)dt$

$$\int_0^3 A'(t)dt = A(3) - A(0) = 24.200 \text{ or } 24.201$$

6-7b

From time $t = 0$ to $t = 3$ years, the cross-sectional area grows by 24.201 square centimeters.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for using Fundamental Theorem of Calculus

1 point is earned for value of $\int_0^3 A'(t)dt$

1 point is earned for meaning of $\int_0^3 A'(t)dt$

$$\int_0^3 A'(t)dt = A(3) - A(0) = 24.200 \text{ or } 24.201$$

From time $t = 0$ to $t = 3$ years, the cross-sectional area grows by 24.201 square centimeters.

6-7b

53.

t (minutes)	0	4	9	15	20
$W(t)$ (degrees Fahrenheit)	55.0	57.1	61.8	67.9	71.0



The temperature of water in a tub at time t is modeled by a strictly increasing, twice-differentiable function W , where $W(t)$ is measured in degrees Fahrenheit and t is measured in minutes. At time $t = 0$, the temperature of the water is 55°F . The water is heated for 30 minutes, beginning at time $t = 0$. Values of $W(t)$ at selected times t for the first 20 minutes are given in the table above.

(a) Use the data in the table to estimate $W'(12)$. Show the computations that lead to your answer. Using correct units, interpret the meaning of your answer in the context of this problem.

(b) Use the data in the table to evaluate $\int_0^{20} W'(t) dt$. Using correct units, interpret the meaning of $\int_0^{20} W'(t) dt$ in the context of this problem.

(c) For $0 \leq t \leq 20$, the average temperature of the water in the tub is $\frac{1}{20} \int_0^{20} W(t) dt$. Use a left Riemann sum with the four subintervals indicated by the data in the table to approximate $\frac{1}{20} \int_0^{20} W(t) dt$. Does this approximation overestimate or underestimate the average temperature of the water over these 20 minutes? Explain your reasoning.

(d) For $20 \leq t \leq 25$, the function W that models the water temperature has first derivative given by $W'(t) = 0.4\sqrt{t} \cos(0.06t)$. Based on the model, what is the temperature of the water at time $t = 25$?

Part A

1 point is earned for the estimation

1 point is earned for the interpretation with units

$$W'(12) \approx \frac{W(15) - W(9)}{15 - 9} = \frac{67.9 - 61.8}{6} \\ = 1.017 \text{ (or } 1.016)$$

The water temperature is increasing at a rate of approximately 1.017°F per minute at time $t = 12$ minutes.



0	1	2
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The student response earns all of the following points:

1 point is earned for the estimation

1 point is earned for the interpretation with units

6-7b

$$W'(12) \approx \frac{W(15) - W(9)}{15 - 9} = \frac{67.9 - 61.8}{6} \\ = 1.017 \text{ (or } 1.016)$$

The water temperature is increasing at a rate of approximately 1.017°F per minute at time $t = 12$ minutes.

Part B

1 point is earned for the value

$$\int_0^{20} W'(t) dt = W(20) - W(0) = 71.0 - 55.0 = 16$$

1 point is earned for the interpretation with units

The water has warmed by 16°F over the interval from $t = 0$ to $t = 20$ minutes.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the value

$$\int_0^{20} W'(t) dt = W(20) - W(0) = 71.0 - 55.0 = 16$$

1 point is earned for the interpretation with units

The water has warmed by 16°F over the interval from $t = 0$ to $t = 20$ minutes.

Part C

1 point is earned for the left Riemann sum

1 points are earned the approximation

1 point is earned for the underestimate with reason

6-7b

$$\begin{aligned}
 \frac{1}{20} \int_0^{20} W(t) dt &\approx \frac{1}{20} (4 \cdot W(0) + 5 \cdot W(4) + 6 \cdot W(9) + 5 \cdot W(15)) \\
 &= \frac{1}{20} (4 \cdot 55.0 + 5 \cdot 57.1 + 6 \cdot 61.8 + 5 \cdot 67.9) \\
 &= \frac{1}{20} \cdot 1215.8 = 60.79
 \end{aligned}$$

This approximation is an underestimate, because a left Riemann sum is used and the function W is strictly increasing.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the left Riemann sum

1 points are earned the approximation

1 point is earned for the underestimate with reason

$$\begin{aligned}
 \frac{1}{20} \int_0^{20} W(t) dt &\approx \frac{1}{20} (4 \cdot W(0) + 5 \cdot W(4) + 6 \cdot W(9) + 5 \cdot W(15)) \\
 &= \frac{1}{20} (4 \cdot 55.0 + 5 \cdot 57.1 + 6 \cdot 61.8 + 5 \cdot 67.9) \\
 &= \frac{1}{20} \cdot 1215.8 = 60.79
 \end{aligned}$$

This approximation is an underestimate, because a left Riemann sum is used and the function W is strictly increasing.

Part D

1 point is earned for the integral

1 point is earned for the answer

$$\begin{aligned}
 W(25) &= 71.0 + \int_{20}^{25} W'(t) dt \\
 &= 71.0 + 2.043155 = 73.043
 \end{aligned}$$

6-7b



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the integral

1 point is earned for the answer

$$\begin{aligned} W(25) &= 71.0 + \int_{20}^{25} W'(t) dt \\ &= 71.0 + 2.043155 = 73.043 \end{aligned}$$